Project Title: Data Analysis of Car Dataset

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Dataset Name: Car Dataset

Kaggle Dataset Link: https://www.kaggle.com/datasets/nehalbirla/vehicle-dataset-from-cardekho?select=Car+details+v3.csv$0

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Purpose of the EDA:-

The main purpose of this Exploratory Data Analysis (EDA) is to understand the used-car dataset and identify important patterns related to car prices, usage, and vehicle characteristics. The analysis includes data cleaning, handling missing values and duplicates, identifying outliers, and using charts to understand relationships between different variables.

The main objective of this project is to analyze a used-car dataset using Python and understand the different factors related to car prices. The project focuses on car details such as manufacturing year, fuel type, transmission, mileage, engine capacity, and kilometers driven.

The analysis includes cleaning the data, handling missing values and duplicates, identifying outliers, and creating visualizations to find useful patterns and relationships. The cleaned data can also be used for further analysis and predicting car selling prices.

Main Questions Selected for Analysis:-

1. What is the distribution of car selling prices?
2. Which fuel types are most common in the dataset?
3. Which transmission type is more common?
4. How are kilometres driven distributed among the cars?
5. Is there a relationship between the manufacturing year and selling price?
6. How are different numerical features related to each other?
7. Are there any unusual values or outliers in the dataset?

Dataset Overview

The dataset used in this project contains information about used cars and their selling prices. It includes details such as the car model, manufacturing year, kilometers driven, fuel type, transmission, mileage, engine capacity, and owner type.

The dataset initially had 8,128 rows and 10 columns. After removing 1,202 duplicate records, 6,926 unique records were available for further analysis.

Feature Description:

Feature	Description	Data Type
name	Name or model of the car	Object
year	Year in which the car was manufactured	Integer
selling_price	Selling price of the car	Integer
km_driven	Total distance covered by the car	Integer
fuel	Fuel used by the car, such as Petrol or Diesel	Object
seller_type	Type of seller	Object
transmission	Type of transmission, such as Manual or Automatic	Object
owner	Previous owner information	Object
mileage	Mileage provided by the car	Object
engine	Engine capacity of the car in CC	Object

Sample Records:

The dataset contains records of different used cars with varying prices, manufacturing years, fuel types, mileage, engine sizes, and kilometers driven. These sample records give us an initial understanding of the type of information available and help us decide which features are useful for further analysis.

```
#First few records
df.head()
```

	name	year	selling_price	km_driven	fuel	seller_type	transmission	owner	mileage	engine
0	Maruti Swift Dzire VDI	2014	450000	145500	Diesel	Individual	Manual	First Owner	23.4 kmpl	1248 CC
1	Skoda Rapid 1.5 TDI Ambition	2014	370000	120000	Diesel	Individual	Manual	Second Owner	21.14 kmpl	1498 CC
2	Honda City 2017-2020 EXi	2006	158000	140000	Petrol	Individual	Manual	Third Owner	17.7 kmpl	1497 CC
3	Hyundai i20 Sportz Diesel	2010	225000	127000	Diesel	Individual	Manual	First Owner	23.0 kmpl	1396 CC
4	Maruti Swift VXI BSIII	2007	130000	120000	Petrol	Individual	Manual	First Owner	16.1 kmpl	1298 CC

Next steps: Generate code with df | New interactive sheet

```
#Last few records
df.tail()
```

	name	year	selling_price	km_driven	fuel	seller_type	transmission	owner	mileage	engine
8123	Hyundai i20 Magna	2013	320000	110000	Petrol	Individual	Manual	First Owner	18.5 kmpl	1197 CC
8124	Hyundai Verna CRDi SX	2007	135000	119000	Diesel	Individual	Manual	Fourth & Above Owner	16.8 kmpl	1493 CC
8125	Maruti Swift Dzire ZDi	2009	382000	120000	Diesel	Individual	Manual	First Owner	19.3 kmpl	1248 CC
8126	Tata Indigo CR4	2013	290000	25000	Diesel	Individual	Manual	First Owner	23.57 kmpl	1396 CC
8127	Tata Indigo CR4	2013	290000	25000	Diesel	Individual	Manual	First Owner	23.57 kmpl	1396 CC

```
#Check number of rows and columns
df.shape
```

```
(8128, 10)
```

```
#Check number of rows
df.shape[0]
```

```
8128
```

```
#Check number of columns
df.shape[1]
```

```
10
```

```
#summary of data
df.info()
```

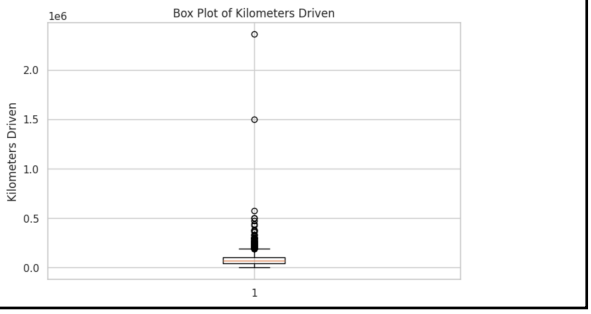
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8128 entries, 0 to 8127
Data columns (total 10 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   name           8128 non-null   object
 1   year           8128 non-null   int64
 2   selling_price  8128 non-null   int64
 3   km_driven      8128 non-null   int64
 4   fuel           8128 non-null   object
 5   seller_type    8128 non-null   object
 6   transmission   8128 non-null   object
 7   owner          8128 non-null   object
 8   mileage        7907 non-null   object
 9   engine         7907 non-null   object
dtypes: int64(3), object(7)
```

```
# Box Plot of Kilometers Driven
plt.figure(figsize=(8,5))

plt.boxplot(df["km_driven"])

plt.title("Box Plot of Kilometers Driven")
plt.ylabel("Kilometers Driven")

plt.show()
```

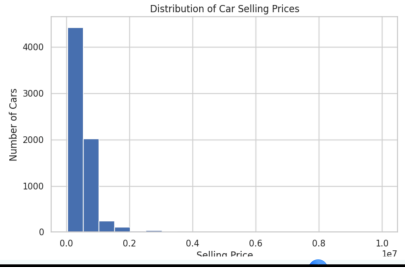



```
# Distribution of Selling Price
plt.figure(figsize=(8,5))

plt.hist(df["selling_price"], bins=20)

plt.title("Distribution of Car Selling Prices")
plt.xlabel("Selling Price")
plt.ylabel("Number of Cars")

plt.show()
```

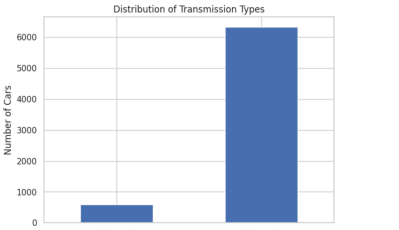



```
# Distribution of Transmission Types
transmission_counts = df["transmission"].value_counts().sort_index()

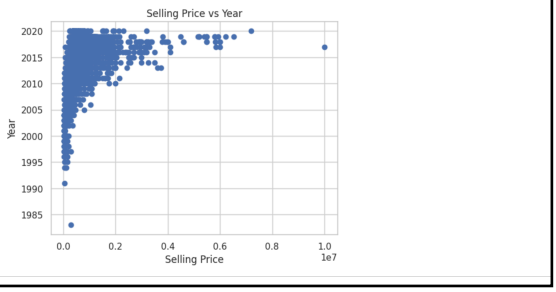
transmission_counts.plot(kind="bar", figsize=(7,5))

plt.title("Distribution of Transmission Types")
plt.xlabel("Transmission Type")
plt.ylabel("Number of Cars")
plt.xticks([0, 1], ["Manual", "Automatic"], rotation=0)

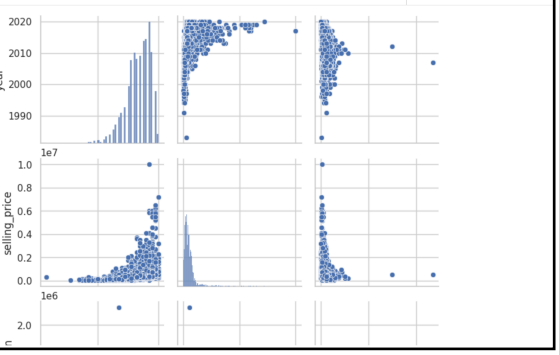
plt.show()
```




```
# Bivariate
import matplotlib.pyplot as plt
plt.scatter(df["selling_price"], df["year"])
plt.title("Selling Price vs Year")
plt.xlabel("Selling Price")
plt.ylabel("Year")
plt.show()
```




```
# Multivariate Analysis
sns.pairplot(df[["year", "selling_price", "km_driven", "mileage", "engine"]])
plt.show()
```

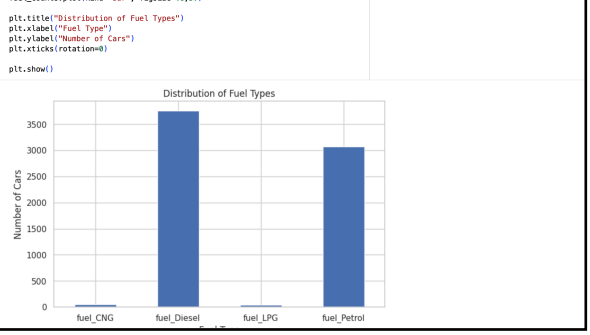



```
# Distribution of Fuel Types
fuel_counts = df[["fuel_CNG", "fuel_Diesel", "fuel_LPG", "fuel_Petrol"]].sum()

fuel_counts.plot(kind="bar", figsize=(8,5))

plt.title("Distribution of Fuel Types")
plt.xlabel("Fuel Type")
plt.ylabel("Number of Cars")
plt.xticks(rotation=0)

plt.show()
```



Observation:

The sample records give a clear idea of the information available for each car, such as its name, manufacturing year, selling price, fuel type, kilometers driven, transmission, mileage, and engine capacity. The graphs also make it easier to understand how these features are distributed and how they are related to each other. Overall, the dataset provides useful information for exploring used-car prices and identifying common patterns in the data.

Data Quality and Cleaning :

Before performing the analysis, the dataset was checked for missing values, duplicate records, outliers, and incorrect data types.

Missing Values

```
[16]
✓ 0s
#check missing value
df.isnull().sum()
```

	0
name	0
year	0
selling_price	0
km_driven	0
seller_type	0
transmission	0
owner	0
mileage	208
engine	208
fuel_CNG	0
fuel_Diesel	0
fuel_LPG	0
fuel_Petrol	0

dtype: int64

The mileage and engine columns contained missing values. These values were handled by converting the columns into numeric format and filling the remaining missing values using the median. This helped retain the available records without unnecessarily removing rows.

Duplicate Record

Check and Remove Duplicate Records

```
#check if any duplicates
df.duplicated().sum()
```

```
np.int64(1202)
```

The dataset contains 1,202 duplicate records. These records have the same values across all the columns and therefore we will not add any new information to the analysis. Keeping these repeated records could affect the accuracy of our results, so they will be removed before continuing with further analysis.

```
#to remove all duplicates
df.drop_duplicates(inplace=True)
```

```
#check if any duplicates
df.duplicated().sum()
```

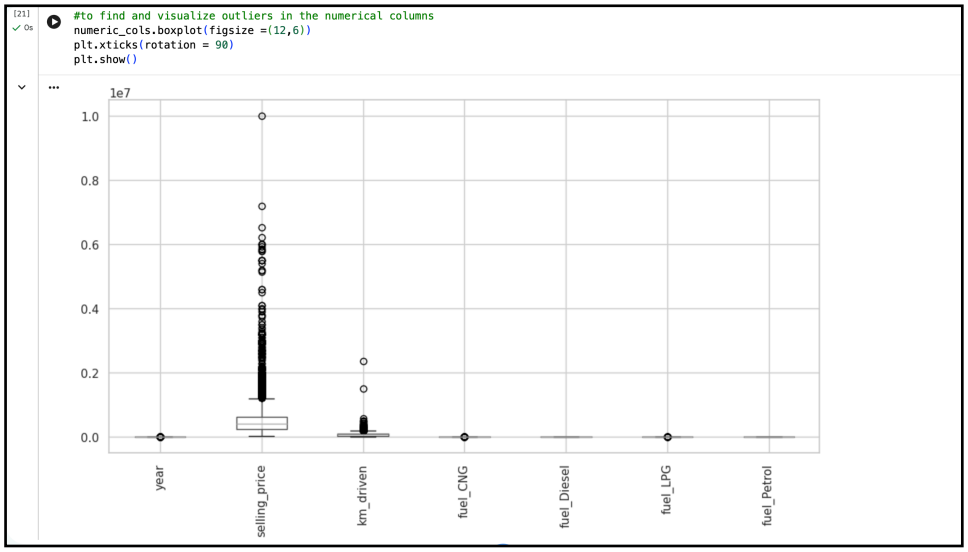
```
np.int64(0)
```

Observation:

After removing the duplicate records, the dataset contains 6,926 unique records and 10 columns. This gives us a cleaner dataset without repeated entries, which will help make the upcoming analysis more reliable.

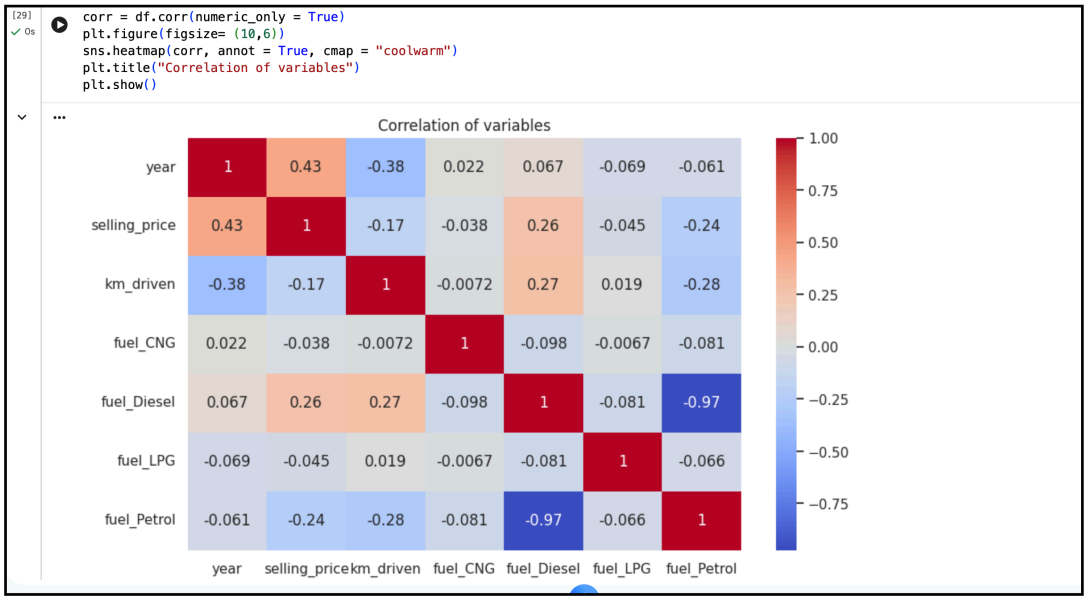
Outliers

Box plots were used to identify unusual values in numerical columns such as selling price and kilometers driven. Some high values were observed, especially in kilometers driven and selling price. These values were retained because they may represent genuine cars rather than errors in the dataset.



Data Type Corrections

The mileage and engine columns were originally stored as text because they included units such as kmpl and CC. The units were removed and the columns were converted into numeric format. Categorical variables such as fuel type, seller type, owner, and transmission were converted into numerical form where required for analysis and machine learning.



Cleaning Decisions

The cleaning process focused on improving the quality of the data while keeping as many original records as possible. Duplicate records were removed, missing values were handled, text-based numerical fields were converted, and potential outliers were reviewed rather than automatically deleted.

Cleaning Sequence

The data cleaning was carried out in the following order:

1. **Dataset Inspection:** The dataset was first checked using head(), shape, and info() to understand its structure, columns, and data types.
2. **Duplicate Check:** Duplicate records were identified using duplicated(). A total of 1,202 duplicates were found and removed.
3. **Missing Value Check:** Missing values were checked using isnull().sum(). Missing values were found in the mileage and engine columns.
4. **Data Type Correction:** The mileage and engine columns were cleaned by removing their units and converting them from text to numeric values.
5. **Missing Value Treatment:** The remaining missing values in mileage and engine were filled using their median values.
6. **Categorical Data Conversion:** Categorical columns such as fuel type, seller type, owner, and transmission were converted into numerical form where required.
7. **Outlier Check:** Box plots were used to identify unusual values in numerical columns. Potential outliers were reviewed and retained when they appeared to be valid observations.
8. **Final Dataset Preparation:** After cleaning, the dataset was checked again to make sure the data types were appropriate and no unwanted missing values remained before starting the analysis and machine learning steps.

Descriptive Analysis:

Numerical Variables

After removing duplicate records, the dataset contains 6,926 unique records.

- The year column shows that the dataset includes both older and newer cars.
- The selling_price values vary widely, meaning the dataset contains cars from different price ranges.
- The km_driven values also vary a lot, showing that some cars have been driven much more than others.
- mileage and engine have different values for different cars, showing differences in fuel efficiency and engine capacity.
- Some unusually high values can be seen in km_driven and selling_price. These were kept because they may be genuine records and not necessarily errors.

Categorical Variables

The dataset also contains categorical information about the cars and their sellers.

- The fuel column includes Petrol, Diesel, CNG, and LPG, with Petrol and Diesel cars forming a large part of the dataset.
- The transmission column contains Manual and Automatic cars, with Manual cars appearing more frequently.
- The seller_type column shows that cars are sold through different types of sellers.
- The owner column contains different ownership categories, which helps us understand the ownership history of the cars.

Overall, the statistics give us a clear picture of the variety of cars in the dataset and provide a useful starting point for studying their prices, usage, and other features.

The summary statistics help us understand the overall range and distribution of the numerical values in the dataset.

- The dataset contains 8,128 car records before removing duplicate records.
- The average manufacturing year is around 2014, with cars ranging from 1983 to 2020.
- The average selling price is about ₹6.38 lakh, but the prices range from around ₹30,000 to ₹1 crore, showing a large difference between low- and high-priced cars.
- The average distance driven is about 69,820 km, while the median is 60,000 km. Some cars have very high driving distances, with the maximum reaching about 23.6 lakh km.
- Most cars fall within a selling price range of approximately ₹2.55 lakh to ₹6.75 lakh, based on the middle 50% of the data.
- For distance driven, 50% of the cars have been driven up to 60,000 km, while 75% have been driven up to 98,000 km.
- The difference between the average and median values suggests that some cars with very high prices or driving distances are affecting the overall averages.

Overall, the dataset shows a good amount of variation in car age, selling price, and distance driven, making these features useful for further analysis.

Key Findings:

- **Key Insights**

1. **Price varies widely:** Selling prices range from ₹30,000 to ₹1 crore.
2. **Mix of old and new cars:** Cars range from 1983 to 2020, with an average year of around **2014**.
3. **Moderate usage:** The median distance driven is 60,000 km, while some cars have much higher values.
4. **Petrol and Diesel dominate:** Most cars use Petrol or Diesel, while CNG and LPG are less common.
5. **Manual cars are more common:** Manual transmission cars are more frequently found than automatic cars.
6. **Newer cars generally cost more:** The analysis shows a general trend of higher prices for newer cars.

- **Limitations**

  - The dataset does not include car condition, accident history, or service history.
  - Some extreme values are present in price and kilometers driven.
  - The dataset comes from a specific source and may not represent the entire used-car market.
  - Important factors such as location, insurance, and additional features are not included.

Conclusion:

Overall Learning

- Learned how to clean and work with a real-world dataset using Python.
- Got better at using Pandas, NumPy, Matplotlib, and Seaborn.
- Learned how to handle missing values, duplicates, incorrect data types, and outliers.
- Learned how to create different graphs and use them to understand patterns in the data.
- Understood how EDA helps in finding useful information from raw data.

Challenges Faced

- Handling missing values in the mileage and engine columns.
- Finding and removing duplicate records.
- Converting values like kmpl and CC from text into numerical values.
- Understanding unusual values with the help of box plots.
- Converting categorical data into numerical form for further analysis and machine learning.

Possible Future Analysis

- Develop a machine learning model to predict car prices.
- Find which features have the most effect on selling price.
- Analyze car brands and models in more detail.
- Compare car prices based on fuel type, transmission, and owner type.
- Try different machine learning models and compare their performance.